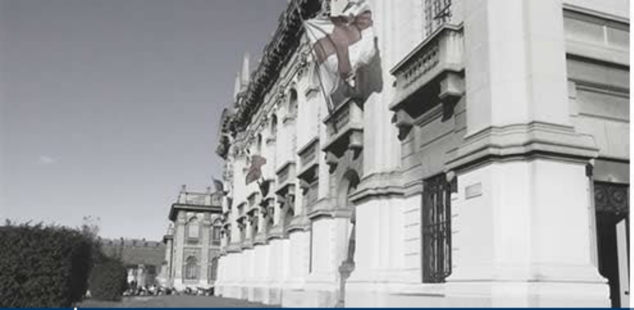

POLITECNICO DI MILANO

Biochip Course

Actuation: Moving Particles

Active Microfluidics

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a.y. 2022/2023

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Forces in Microfluidic Systems

Forces driving fluids:

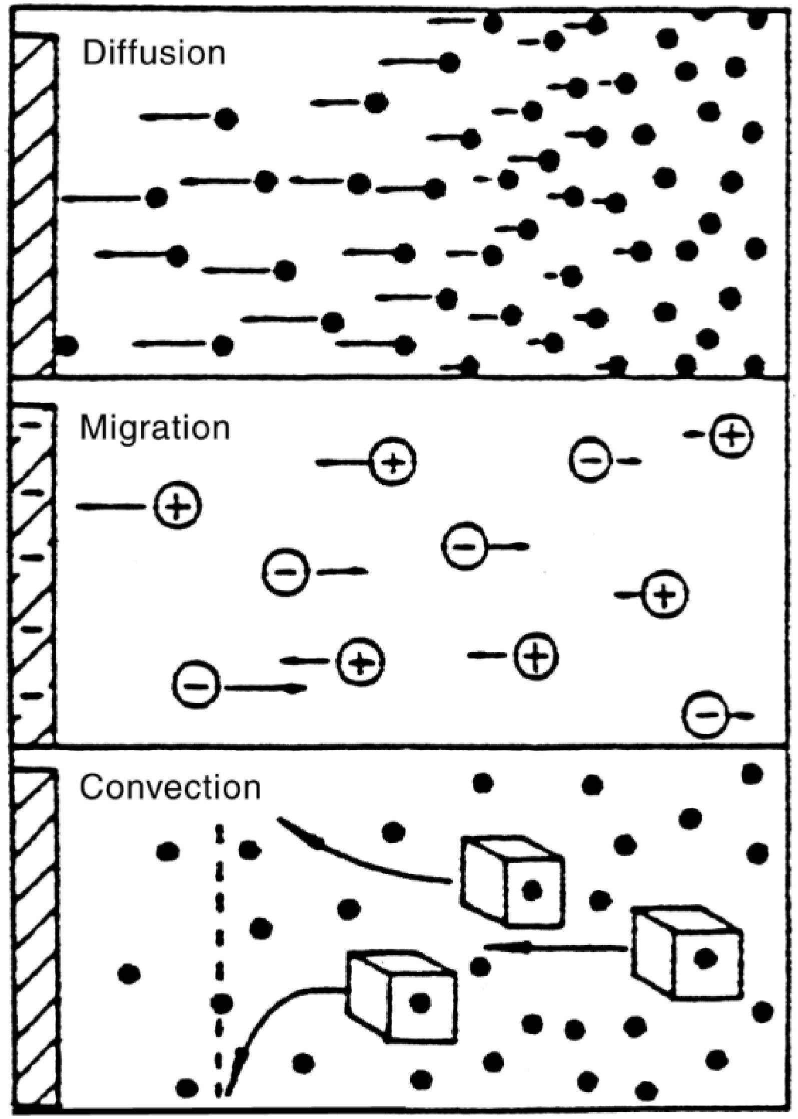
- Hydraulic pressure
- Electro-osmosis (for electrolytes)

Forces driving particles and molecules inside fluids:

- Diffusion (concentration gradient)
- Electrophoresis (charged particles)
- Dielectrophoresis (dielectric particles)
- Magnetic forces (magnetic particles)
- A different paradigm: droplet-based microfluidics

Molecules Transport Mechanism

- **Diffusion**
- **Migration (drift)**
- **Convection (fluid motion)**
  - Natural (density gradient)
  - Mechanical (stirring, flow in microfluidic channel…)



Diffusion

A spatial gradient of molecule (mass) concentration C [kg/m^3] produces a flux:

Fick's law

$$\boxed{J = -D\frac{\partial C}{\partial x}}$$

D = diffusion coefficient [m^2/s]

$$J = \frac{\delta m}{\delta t\ \delta A}$$

$$\boxed{L_{diff} = \sim\sqrt{Dt}}$$

Brownian motion tends to the equilibrium with a random walk

Diffusion Coefficient

For spherical particles of radius *r* in fluids at low *Re*:

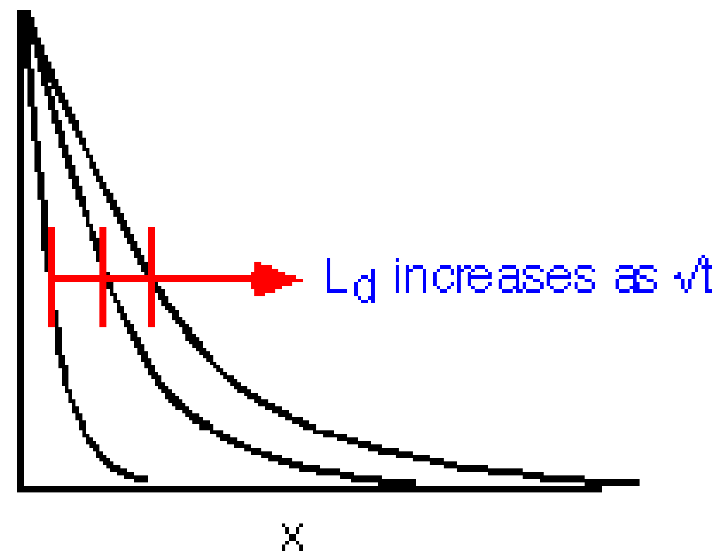
$$\boxed{D = \frac{kT}{6\pi\eta r}}$$

For charged particles:

$$D = \frac{\mu kT}{q}$$

size

Solute	D (cm^2/s)
water	10^{-5}
fluorescein	3×10^{-6}
sucrose	4.6×10^{-10}
myosin	1.2×10^{-11}



Diffusion Times

For the K^+ ion, $D = 2 \cdot 10^{-5}$ cm^2/s

to cover: L = 1cm $\rightarrow$ Δt = 7 hours

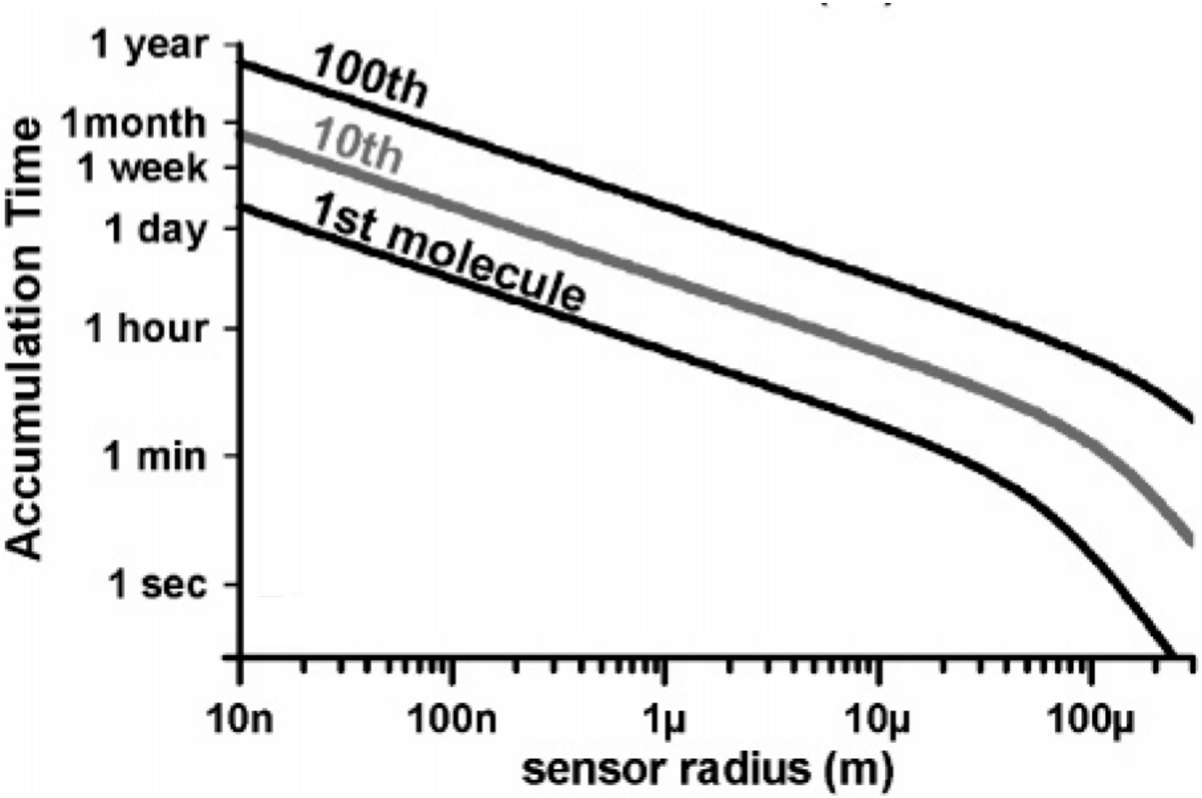
L = 10µm $\rightarrow$ Δt = 25 ms

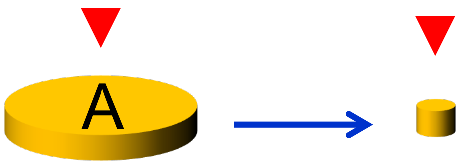
Miniaturization helps reducing the diffusion time!

However, diffusion is a slow mechanism and in order to bring analyte molecules, macromolecules or particles in the detection area (for instance in contact with a sensing electrode) it is preferable to use other forces (active solutions).

Single Molecule Detection: an Extreme Example

Reducing the sensor area, the relative surface variation $\Delta A/A$ increases, but relying on diffusion only causes and increase of the deposition time:





1fM ($D=1.5\ 10^{-6}\ cm^2/s$) (1 molecule in a volume of 120µm x 120 µm x 120µm)

Whitman, "Detection limits for nanoscale biosensors", *Nano Letters*, 5, p. 804 (2005)

Peclet Number

Dimensionless ratio between advection and diffusion:

$$Pe = \frac{v \cdot L}{D}$$

v = fluid velocity [m/s]

L = length scale [m]

D = diffusion coefficient [m^2/s]

Quantitative figure of merit of the competition between forced flow and diffusion transport (similar to Reynolds number, but in this case there is not a narrow threshold range).

Jean Claude Peclet 1793 -1857

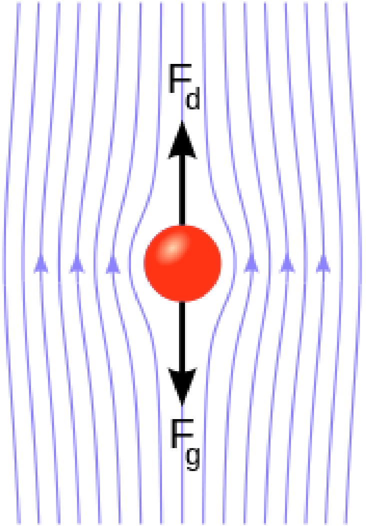
Drag Force

A particle moving in a viscous fluid (viscosity η) with relative velocity v, experiences a **friction force (drag)**. For spherical particles of radius *r* and at low *Re* (inertia neglected):

$$\boxed{F_{drag} = -6\pi \cdot \eta \cdot r \cdot v}$$

Stoke's law (valid as long as *r* >> size of fluid molecules).

Combined with Gravity, Archimedes' force and electrokinetic forces, they set the particle movement.



Electrokinetics

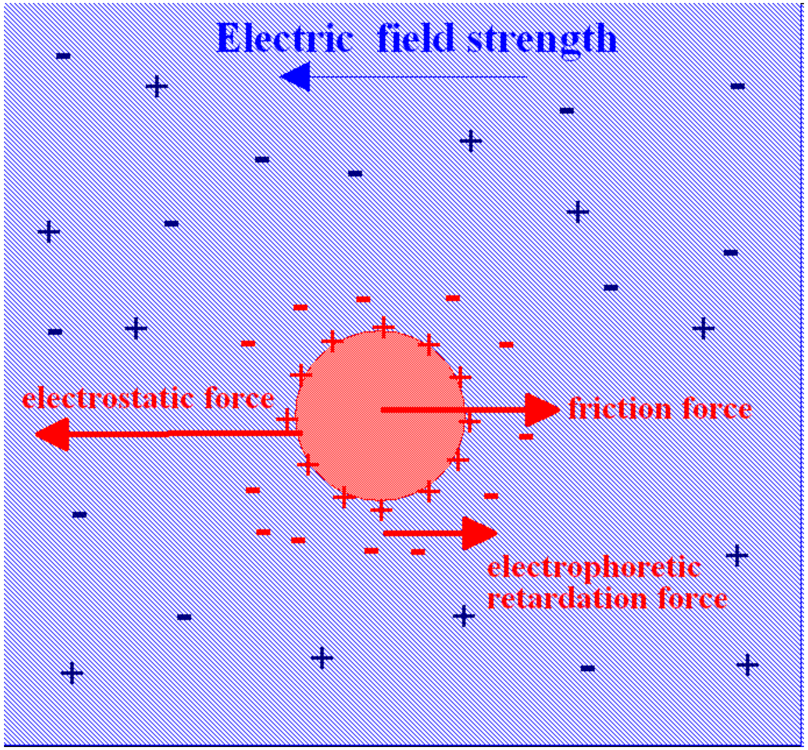
Electrostatic Coulomb force on a charged particle (q) immersed in an electric field E, in an insulating liquid:

$$F_E = qE = F_{Drag} = 6\pi r\eta v$$

Migration velocity:

$$\boxed{v = \frac{q}{6\pi r\eta}E = \mu E \quad \rightarrow \quad \mu = \frac{q}{6\pi r\eta}}$$

Electrophoretic mobility



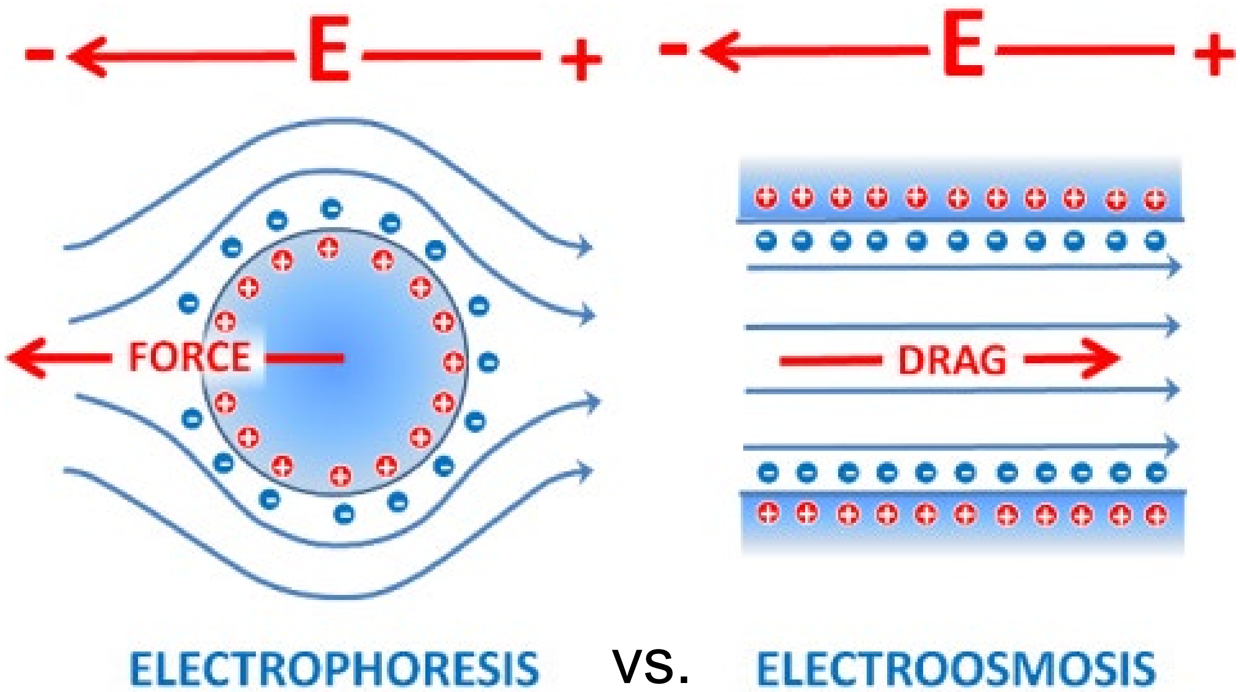
If the liquid contains ions (electrolyte) their charge will shield the particle charge (forming a double layer), thus reducing this effect.

Electrophoresis

The electric field can be used to manipulate **charged** particles.

Electrophoretic techniques can be used both for particle transport and analysis (separation, sizing, identification).

Flow comparison:

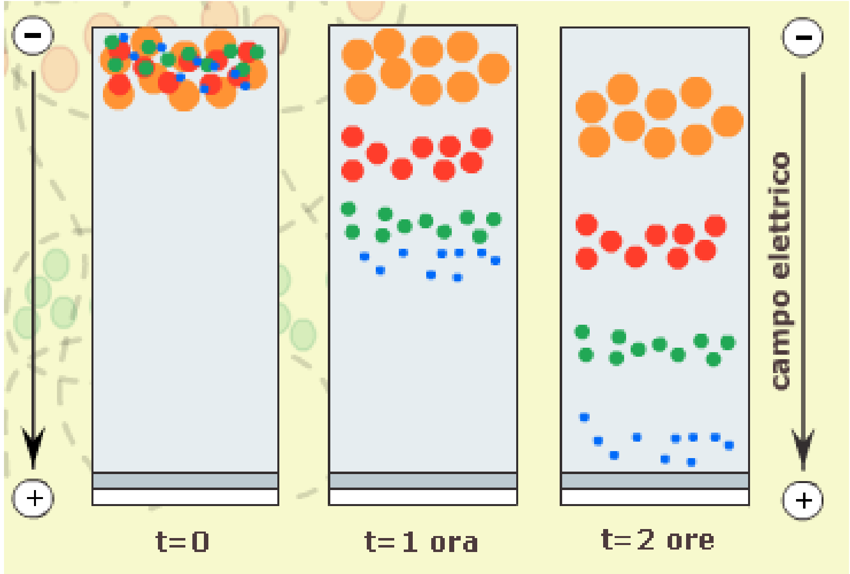


Electrophoretic separation

In a constant E, the transit time of the molecule depends on its mobility

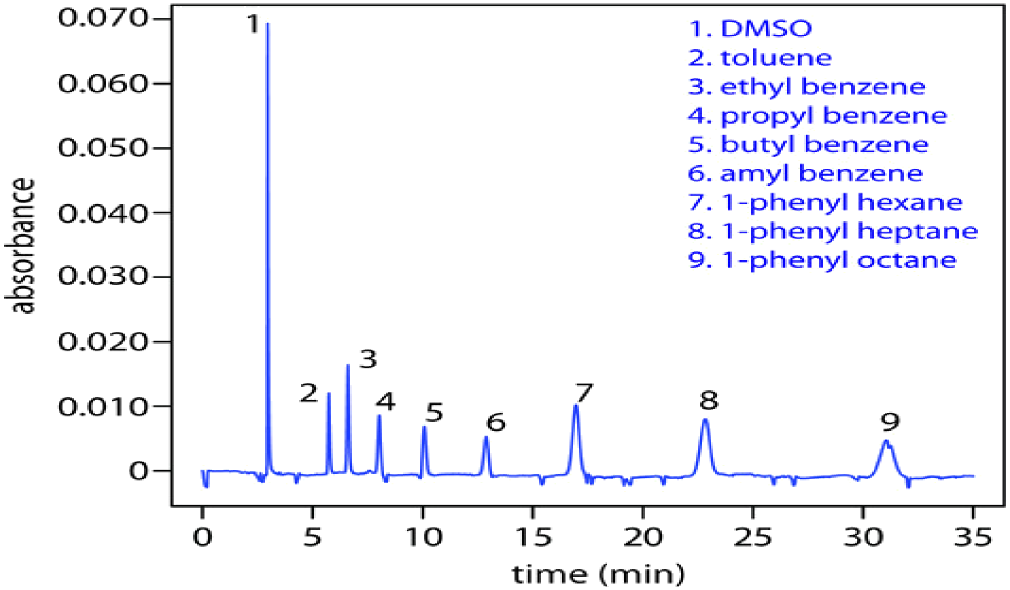
$$\mu \sim q/r$$

i.e. depends on the size



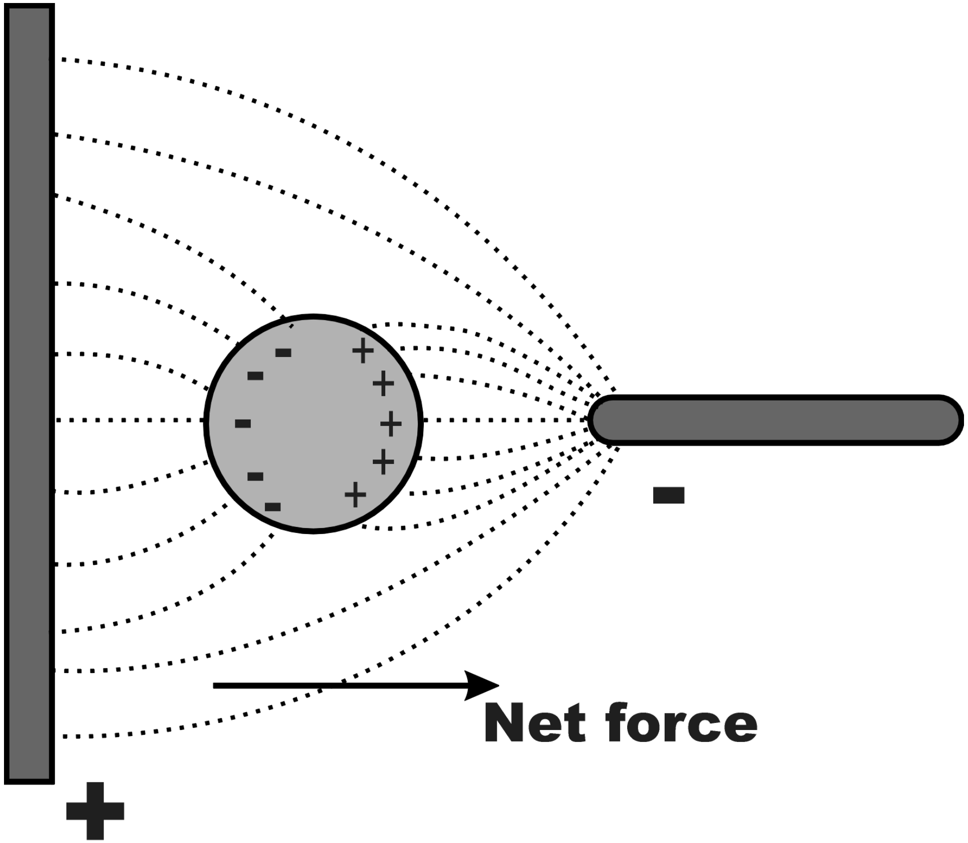
Ion/molecule identification is performed:

- in capillars
- in gel



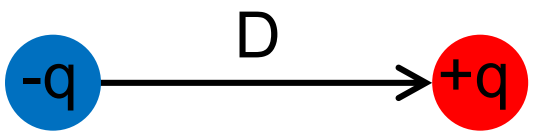
Dielectrophoresis (DEP)

Neutral dielectric particles in liquids can be moved by means of *dielectrophoresis* (named in 1978 by Pohl) a net force that acts on the particle in a non-uniform electric field (gradient):



Polarization

Given an electric dipole:



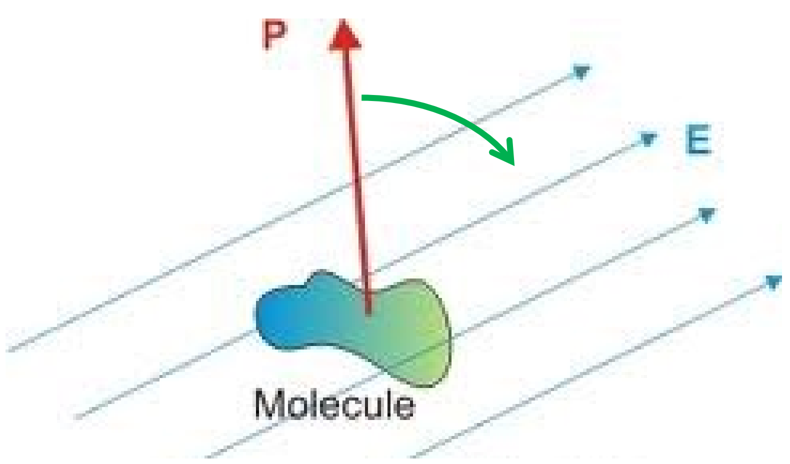
The *Electric Dipole Moment* is: $\vec{P} = q\vec{D}$ [C·m]

D is the displacement vector (pointing from –q to +q)

Polarizability is the ability of the particle to be polarized in an external field E, i.e. the ratio between the induced dipole moment P and E:

$$\vec{P} = \alpha\vec{E} \qquad \alpha = \frac{\vec{P}}{\vec{E}} \qquad [\mathrm{C\cdot m^2/V}]$$

The dipole aligns with E



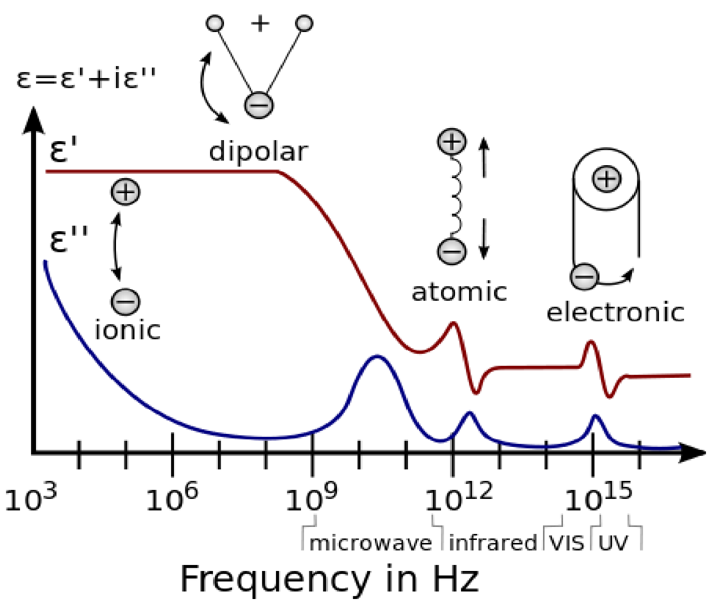
Complex Permittivity

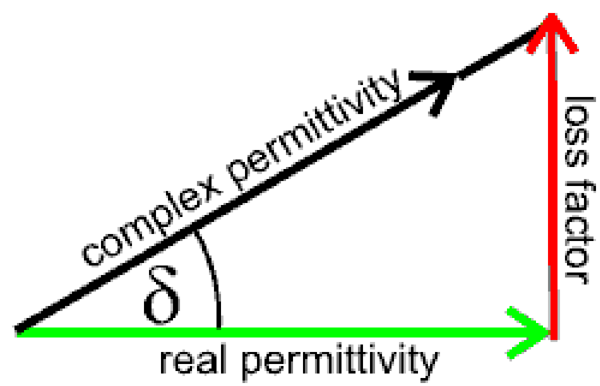
The electric field can be applied in AC at frequency ω.

To study the DEP effect as a function of frequency, it is necessary to use the **complex permittivity**:

$$\varepsilon^* = \varepsilon + \frac{\sigma}{j\omega}$$

where σ is the conductivity (related to loss)





Dielectrophoretic Force

The force is proportional to the field gradient:

$$\vec{F} = \vec{P} \cdot \nabla E = \alpha \nabla E^2$$

for a homogenous spherical particle (radius r, permittivity ε_p) in a surrounding medium ε_m the force is:

$$\boxed{\vec{F} = 2\pi \cdot r^3 \cdot \varepsilon_m \cdot Re\{K_{CM}(\omega)\} \cdot \nabla E^2}$$

- is independent of the field sign
- scales with the volume (r^3)
- changes with frequency

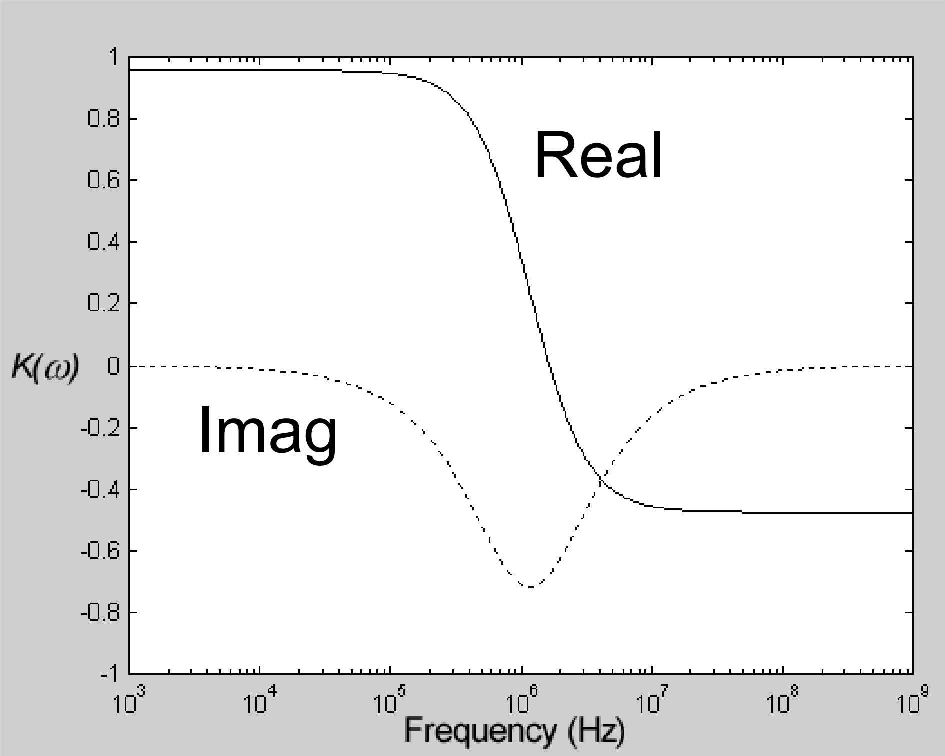
The Clausius-Mossotti Factor

$$K_{CM}(\omega) = \frac{\varepsilon_p{}^* - \varepsilon_m{}^*}{\varepsilon_p{}^* + 2\varepsilon_m{}^*}$$

It is a complex frequency-dependent coefficient describing the relation between the polarizability and the particle and medium permittivity

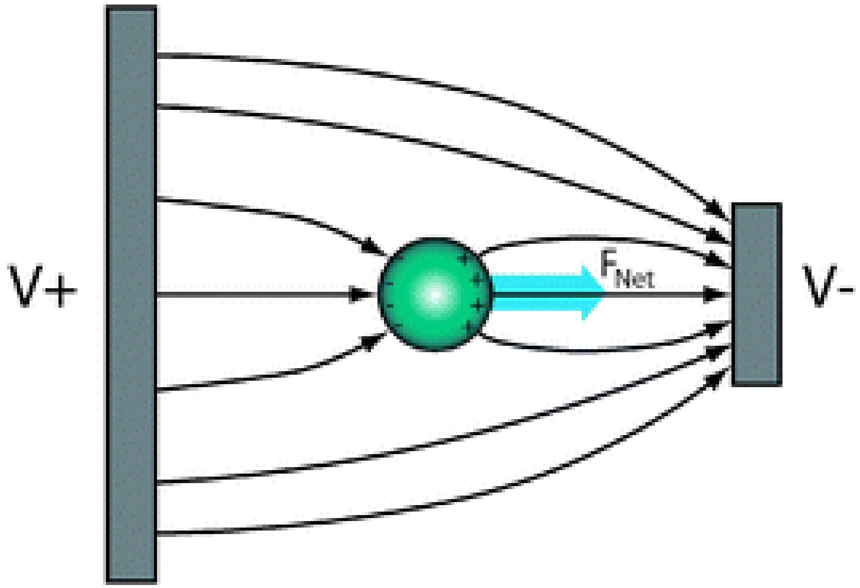
Re → DEP force

(Im → Electrorotation)



Force Direction

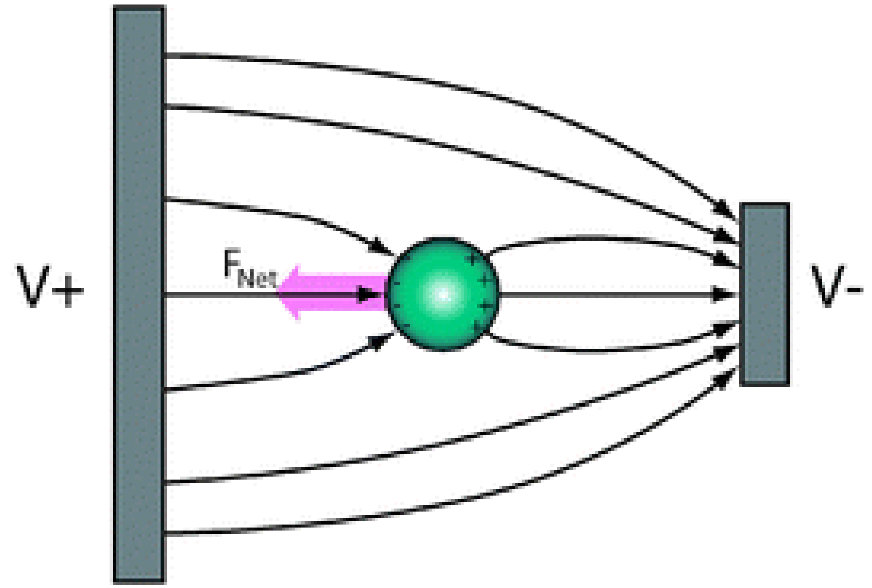
Positive (pDEP)



$\varepsilon_p > \varepsilon_m$

The particle moves toward the maximum of field

Negative (nDEP)



$\varepsilon_m > \varepsilon_p$

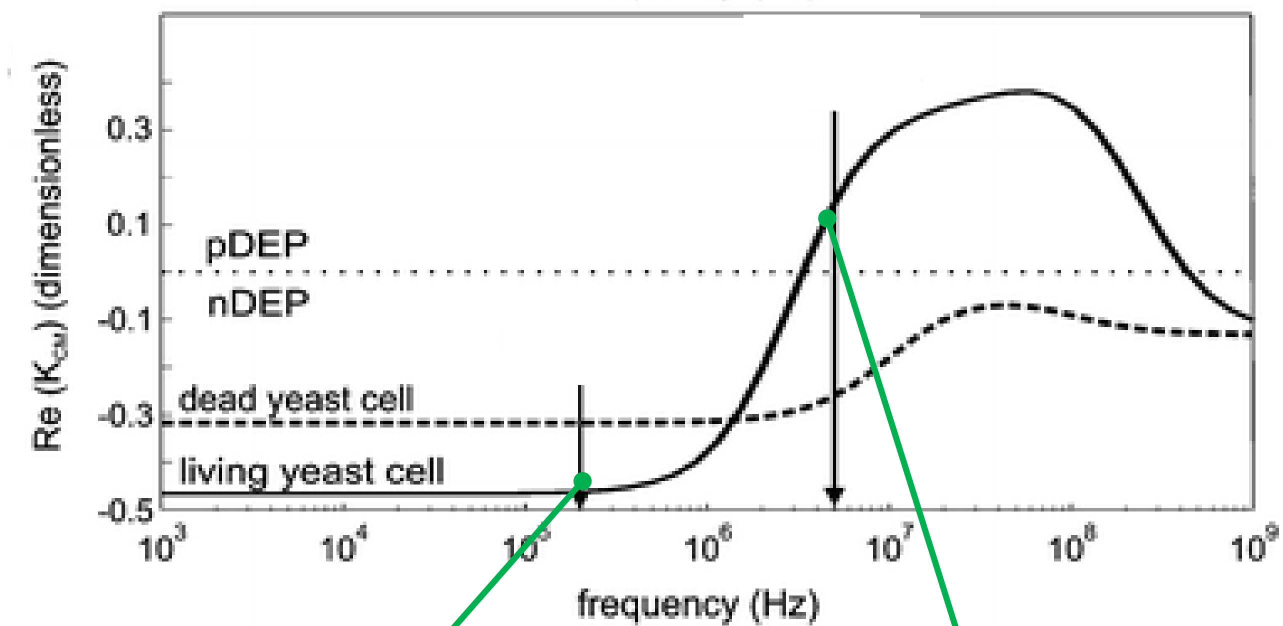
Polar liquid more polarizable than the particle

The particle is moved toward the minimum of field

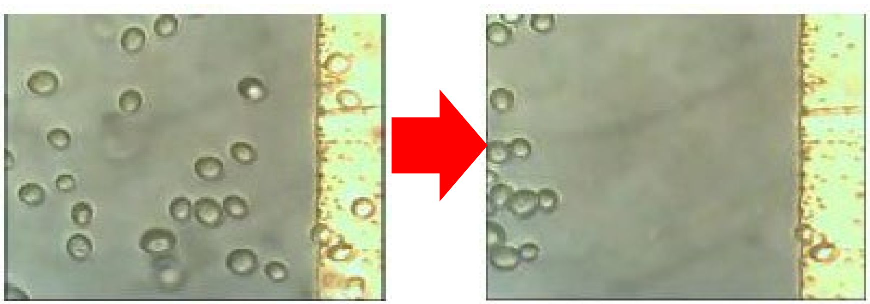
Manipulation of Yeast Cells

Selective frequency

Discriminate different cells (live vs dead)

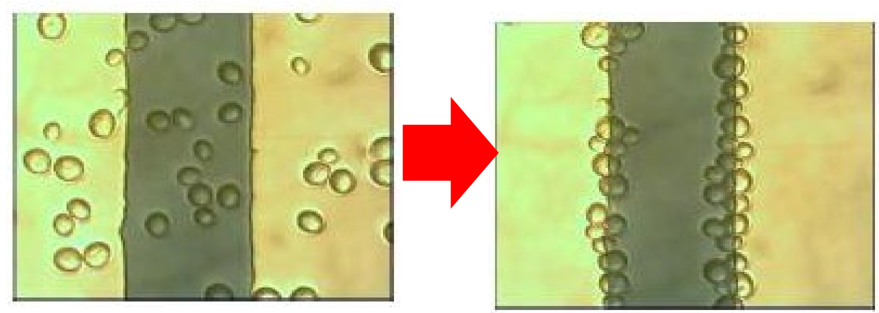


Repulsion (nDEP 200kHz)



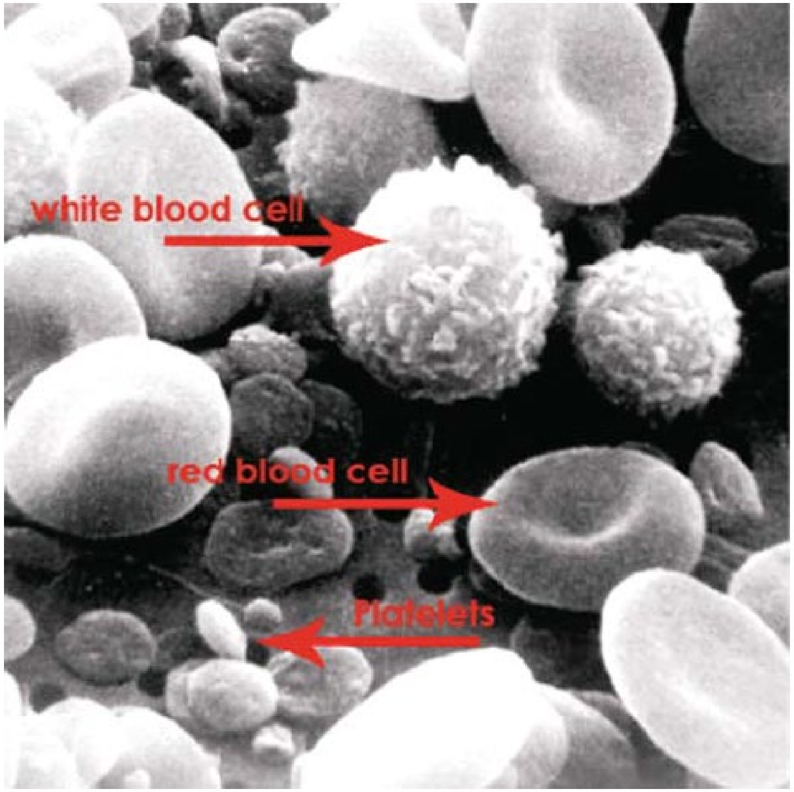
Off On

Attraction (pDEP 5MHz)



Off On

Applications: HIV Detection Example



Goal: count the number of White Blood Cells expressing the CD4 membrane protein.

HIV virus reduces the number of CD4+ T lymphocytes.

In 1 µl of blood:

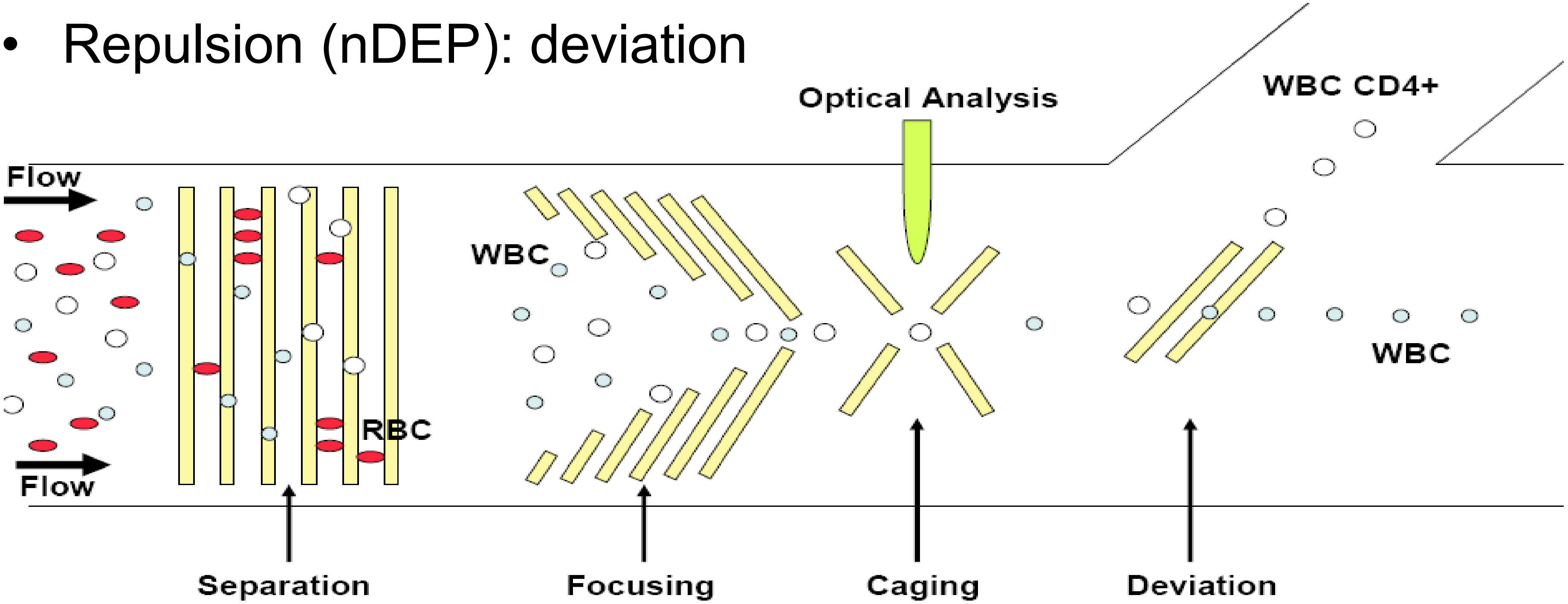
- healthy patient: 700-1000
- HIV infected < 200

Solution: **a single LOC based on DEP**

HIV Example: Functions

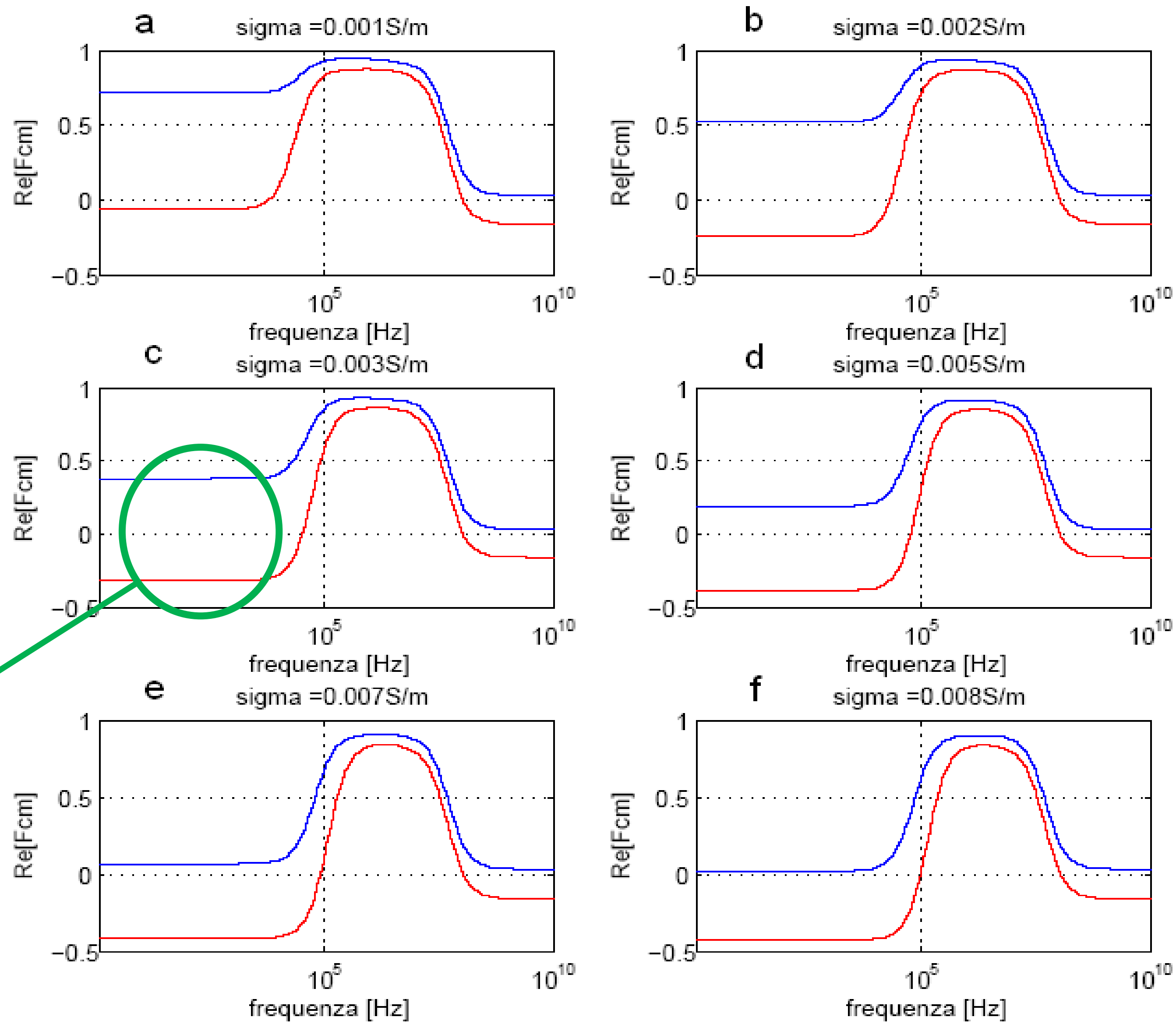
Cascade of functions:

- Trapping (pDEP): separation
- Focusing (nDEP): optical detection of fluorescent CD4 markers
- Trapping: for optical detection
- Repulsion (nDEP): deviation



DEP Separation

Adjust frequency and medium conductivity

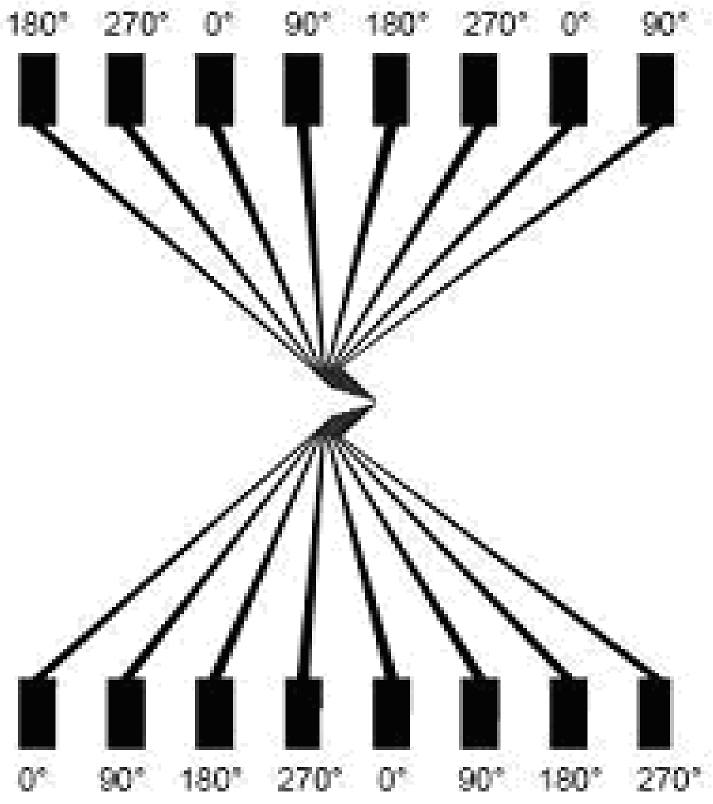


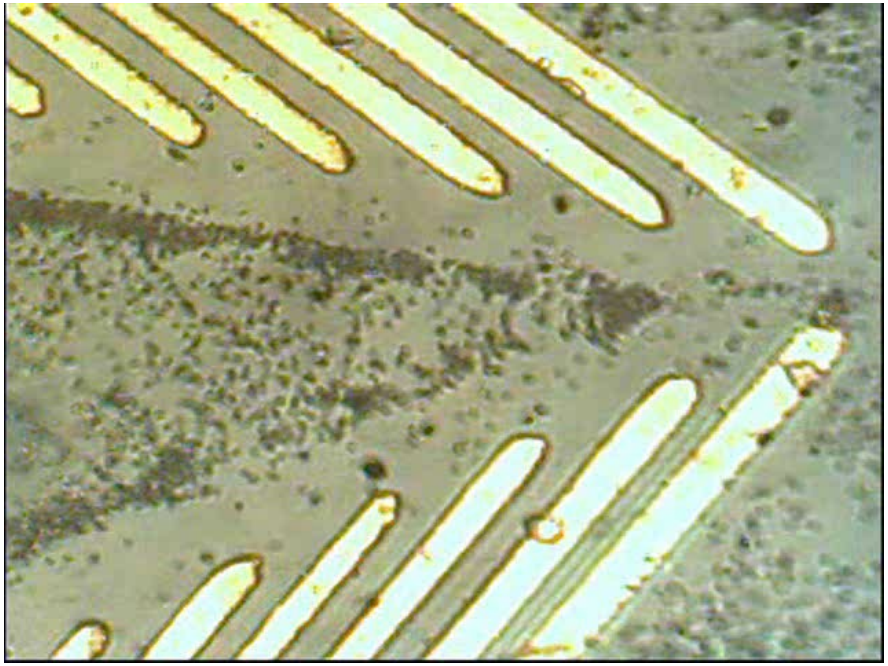
nDEP for WBC

pDEP for RBC

DEP Focusing

Repulsion is preferred instead of attraction for dynamic steering in order to avoid cells/particles sticking

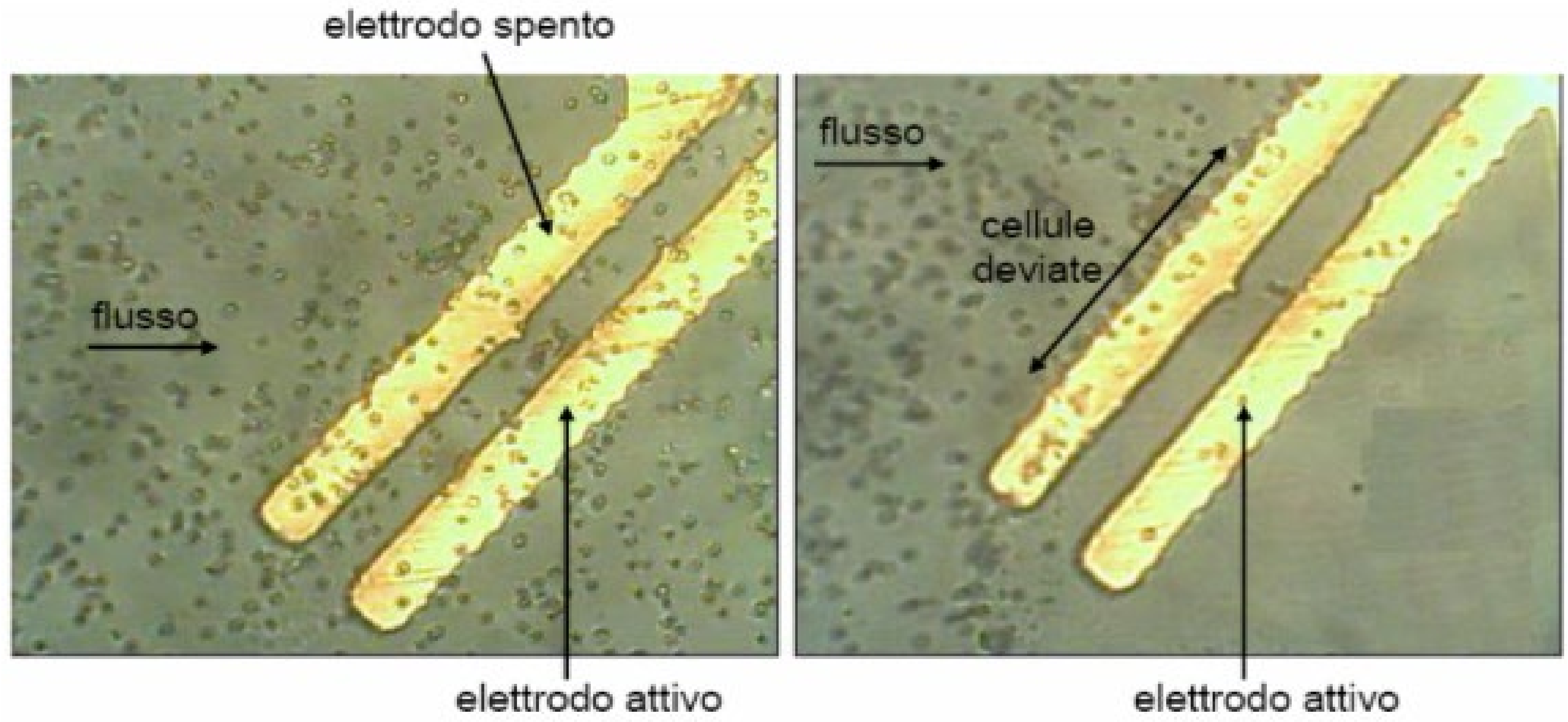




DEP Controlled Deviator

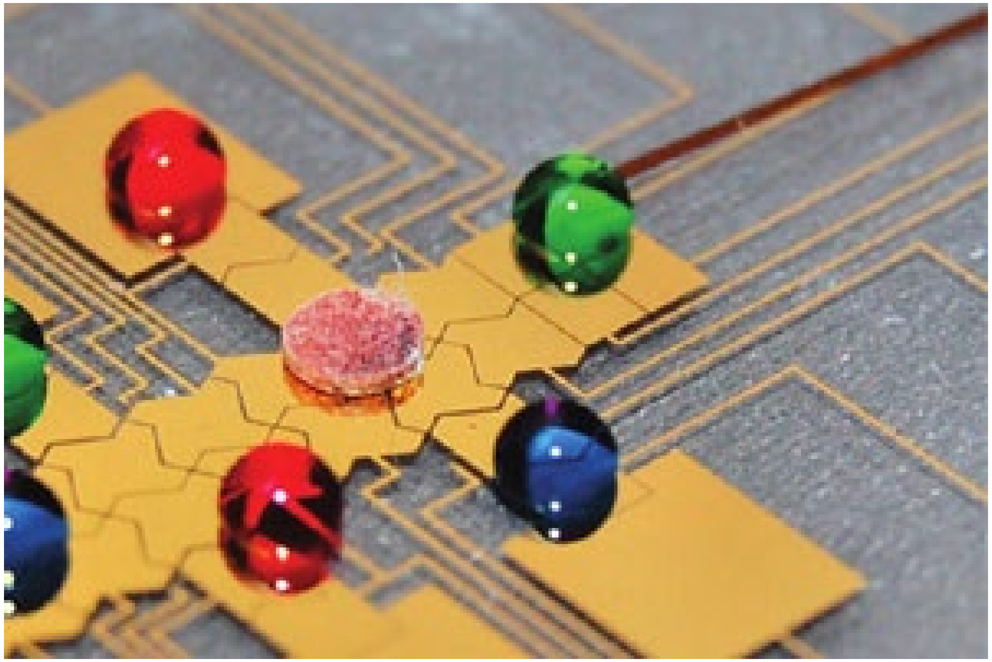
Repulsive barrier activated by the microscope after fluorescence detection

OFF **ON**



Digital Microfluidics

Alternative recent approach (with respect to the continuous flow) based on the manipulation of **single droplets**.



Different fluidodynamics

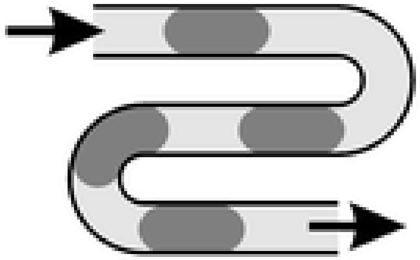
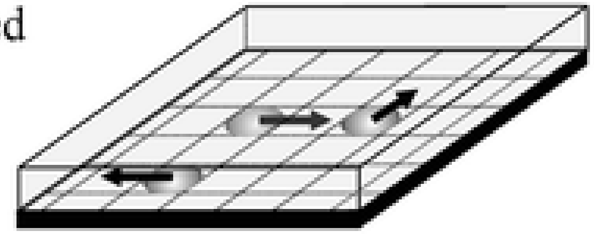
Advantages:

- Well defined «units» of liquid and samples
- Precise and arbitrary spatio-temporal control

Drawbacks:

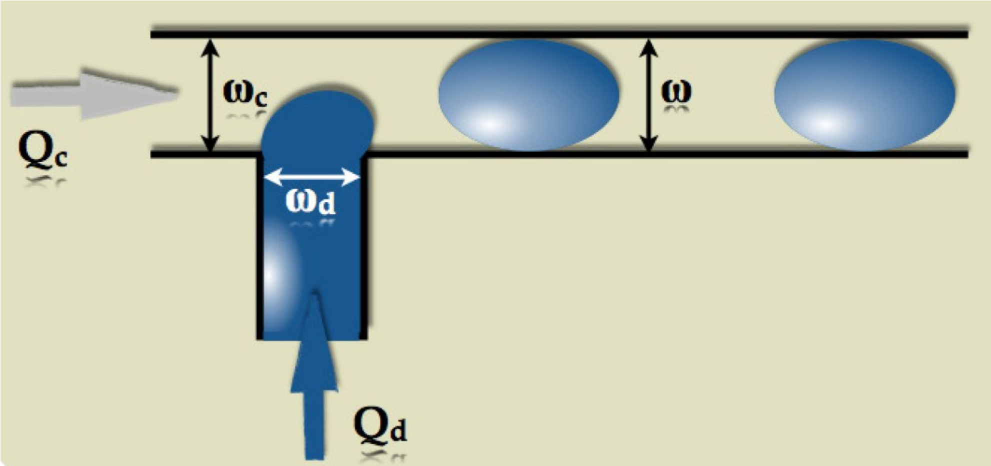
- In air, risk of evaporation
- More complex control electronics (rounting, connections…)

Implementation Approaches

approach	schematic sketch	description
channel based **1.**		- liquid flow driven actuation (pressure driven) - simultaneous motion in 1 dimension - droplet movement defined by geometry - closed microchannels
surface based **2.**		- actuation by SAW or EWOD - individual motion in 2 dimensions - arbitrary droplet movement on surface - planar, (open) surface - online control of assay protocol

1. Droplet Generation

Continuous flow



Discrete flow

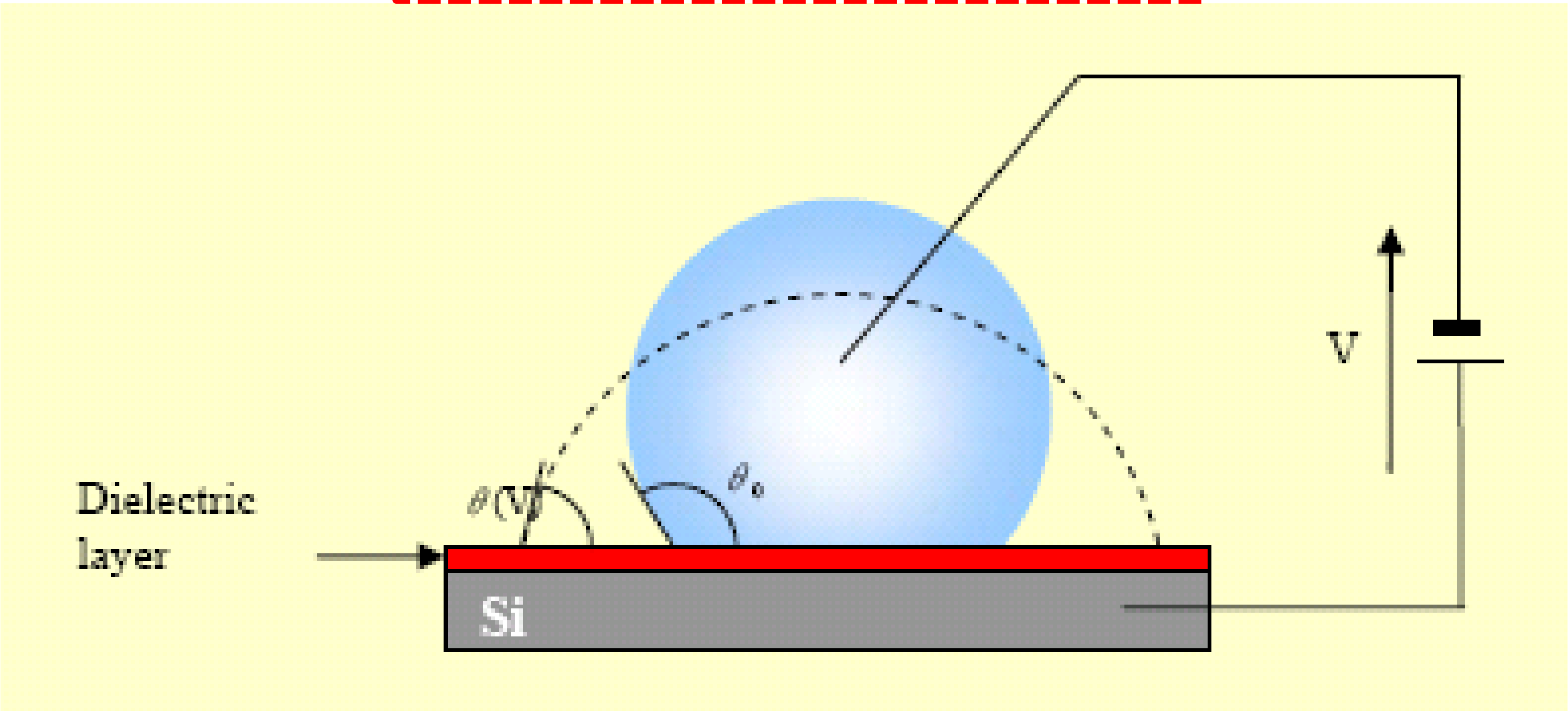
Use of two immiscible fluids (for instance, water and oil).

The size of the droplets is easily controlled by the flow rates.

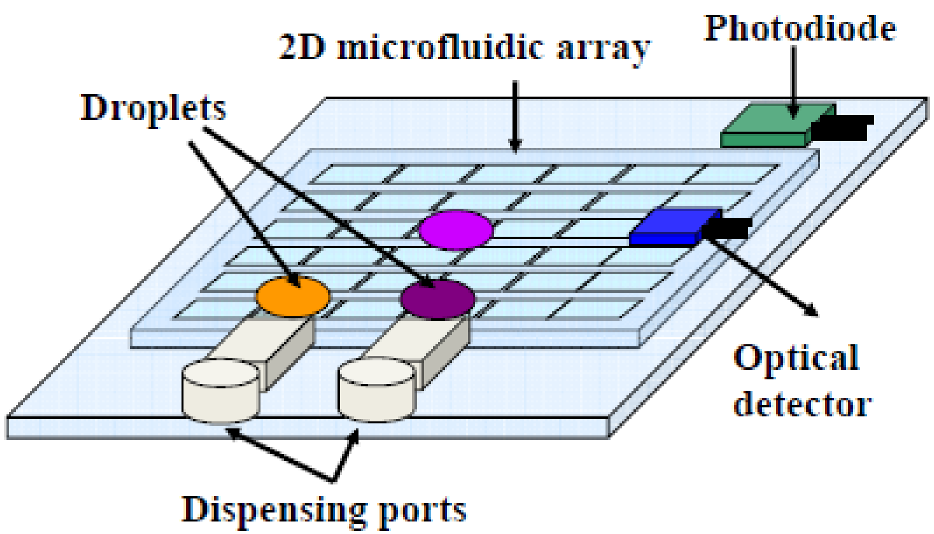
2. Electrowetting

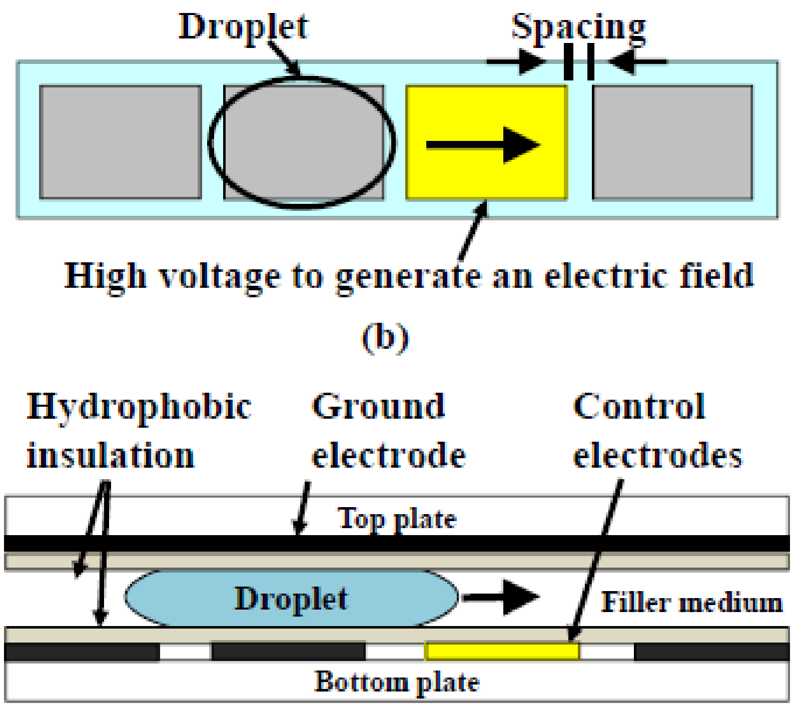
Electric control of the contact angle: the dielectric layer is hydrophobic and when the voltage is applied, the capacitance is charged and electrostatic forces attract the liquid

$$\boxed{cos(\theta) = cos(\theta_0) + \frac{1}{2\gamma} CV^2}$$



2. Electrowetting Approach

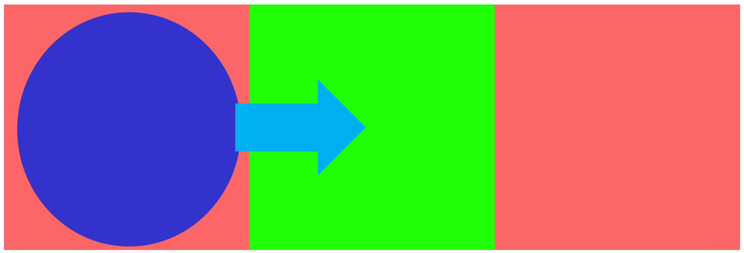


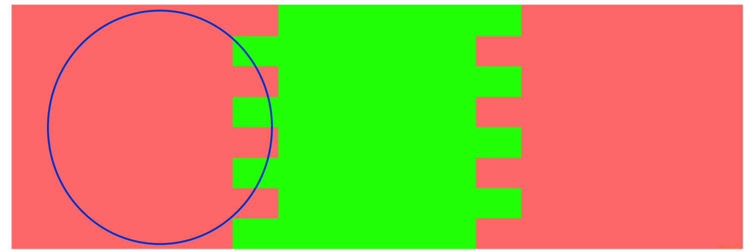


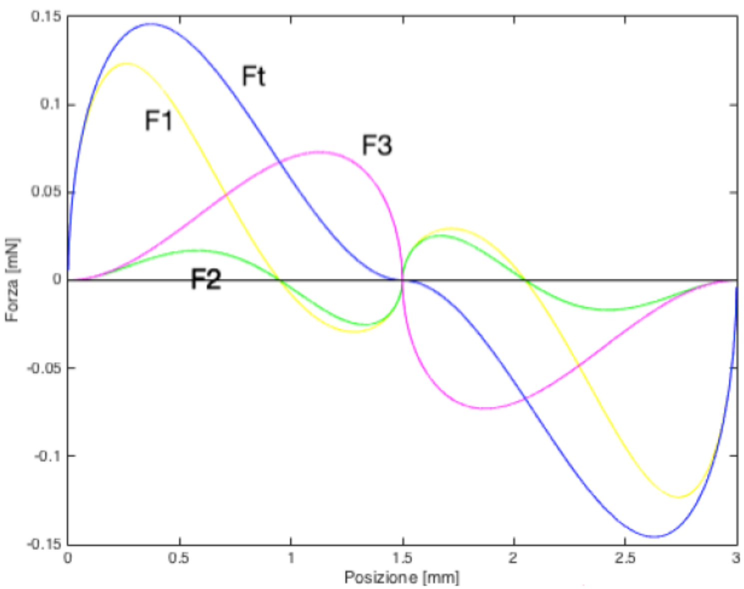
Large arrays of electrodes defining the transport pattern to be driven with a multichannel driving/control electronics

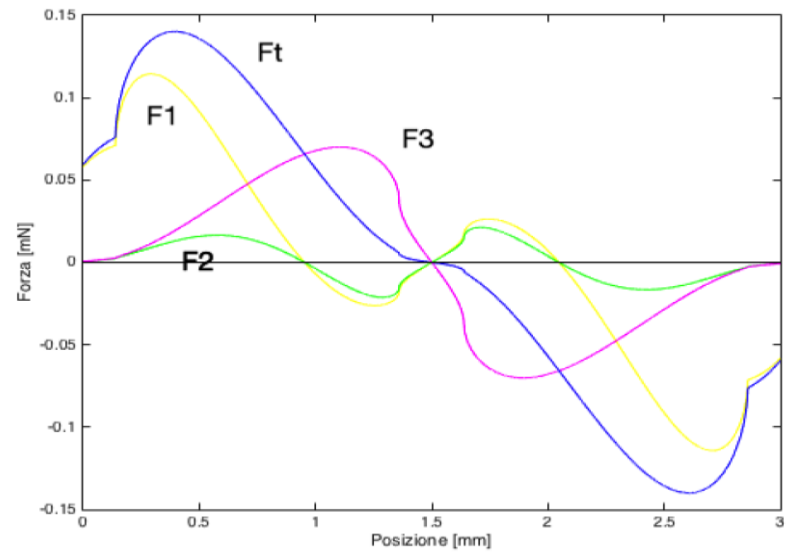
Electrodes Design

Need for partial overlap between adjacent pads to have lateral force







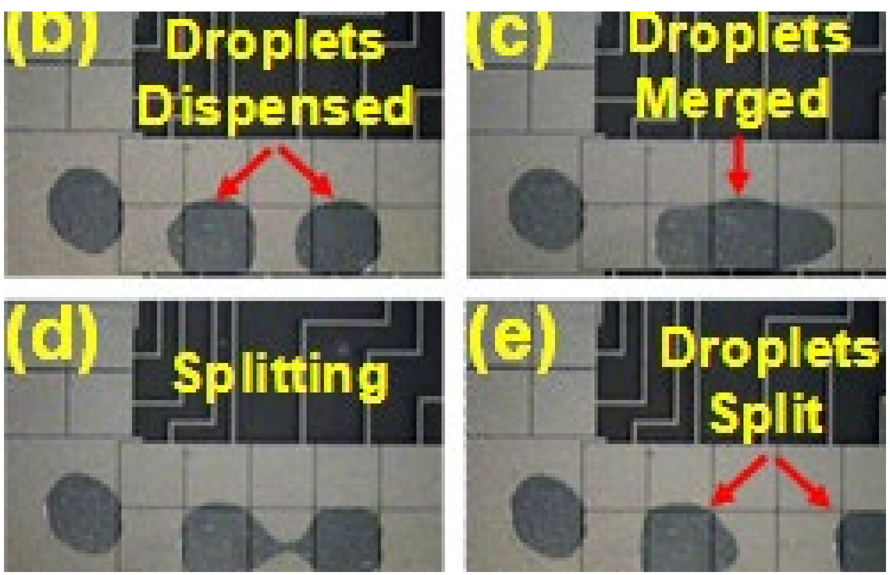


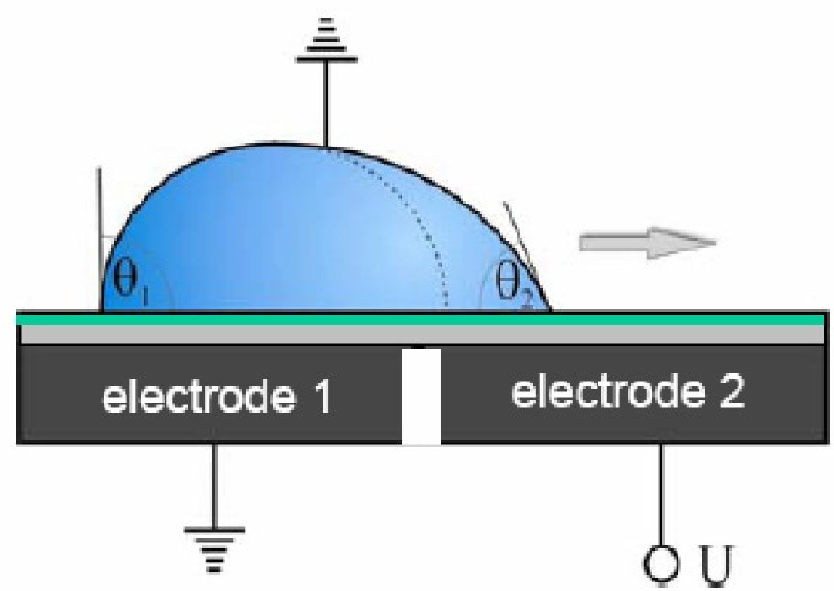
Interdigitated borders

Functions: Spatial Patterns

- Droplet generation
- Droplet transport
- Droplet splitting
- Droplet merging
- Droplet mixing

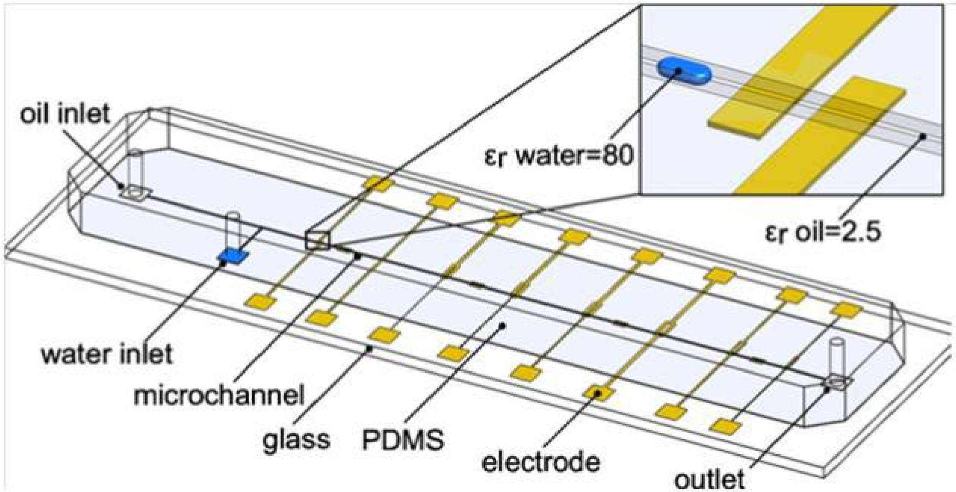
Spatial electric field gradient for lateral displacement

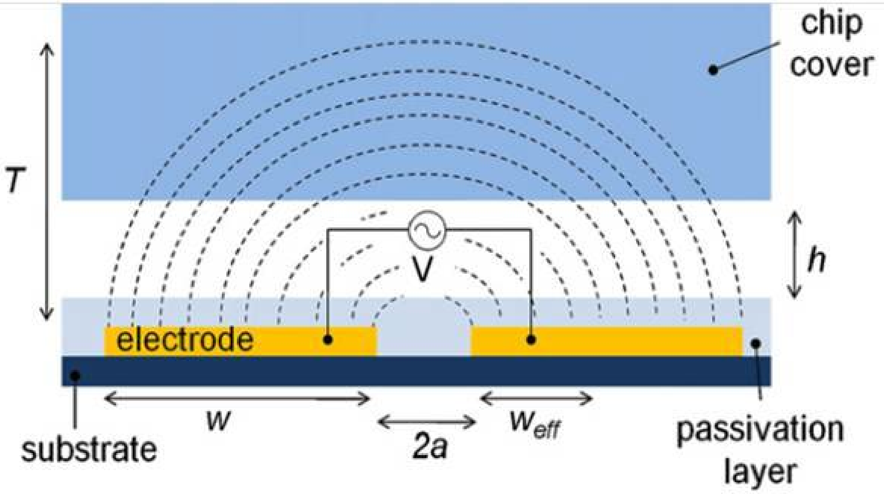


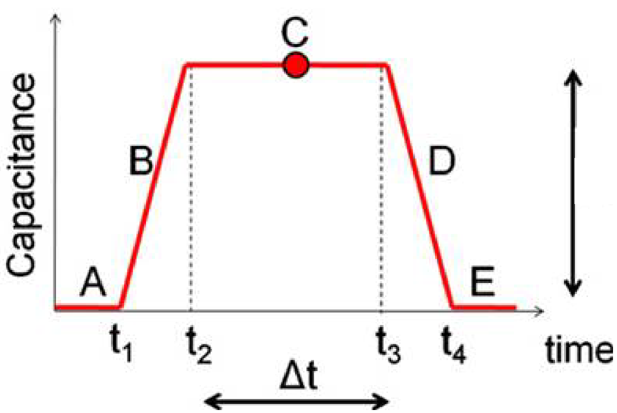


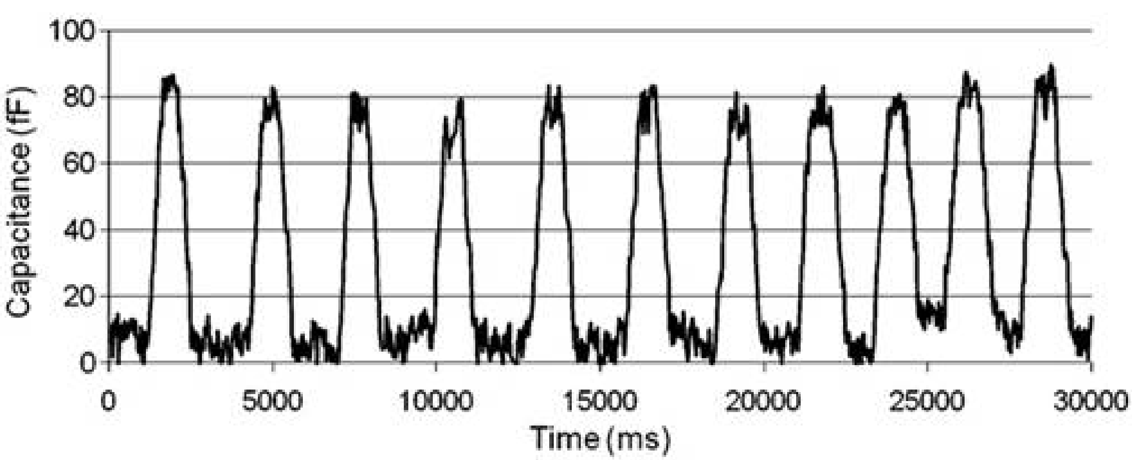
Suitable for Capacitance Detection of Droplets

Water/oil or water/air large dielectric contrast for detection









Evaporation

Evaporation of a droplet: $d^2 = {d^2}_0 - \beta t$

Diameter   Initial diameter   Independent of size

Complete evaporation time: $time = \frac{{d_0}^2}{\beta}$ scales with size^2

If the droplet contains molecules, the evaporation of the solvent produces and increase of their concentration.

It the droplet is an electrolyte, the concentration increases the conductivity.

Key Take-Home Messages

- Instead of moving the fluid, it is possible to directly move the particles inside the fluid.
- Particles moving in a fluid are subject to the *drag* force.
- Diffusion is a slow process (miniaturization speeds-up).
- Charged particles can be moved by *electrophoresis*.
- Electrophoresis is used to separate and identify ions and molecules with different mobility.
- Neutral dielectric particles can be moved by *dielectrophoresis*.
- Dielectrophoretic force changes with frequency and can be attractive or repulsive (trapping or barrier for multiple applications).
- Droplet-based microfluidics (based on *electrowetting*) is a viable alternative to continuous flow.